Intelligent Robotics : Manipulator Control

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Introduction

- Linear control: inertia seen by each joint actuator is approximately constant
- Nonlinear manipulator dynamics
- Approximate method

- Computed torque method
- Lyapunov's stability theory

Nonlinear and time-varying systems

- Local linearization about operating point, moving linearization
- It uses a nonlinear control term to cancel a nonlinearity in the system, so that the overall closed loop system is linear
- Outer control loop : linear control

Spring-Mass System W/O Friction

$$K = \frac{1}{2}m\dot{x}^2 \quad V = \frac{1}{2}kx^2$$

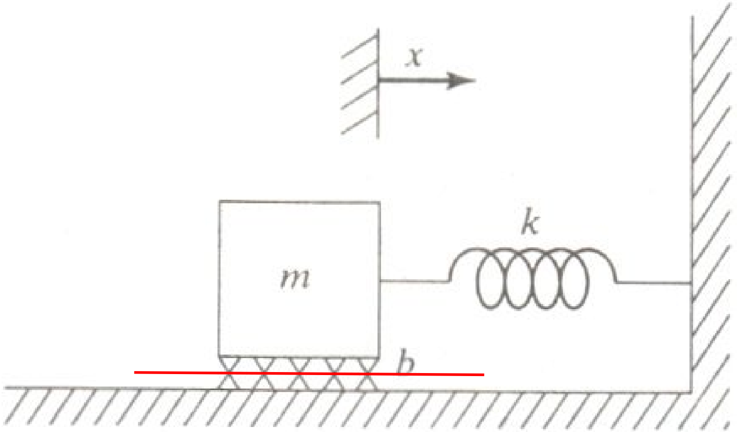
$$m\ddot{x} + kx = 0$$

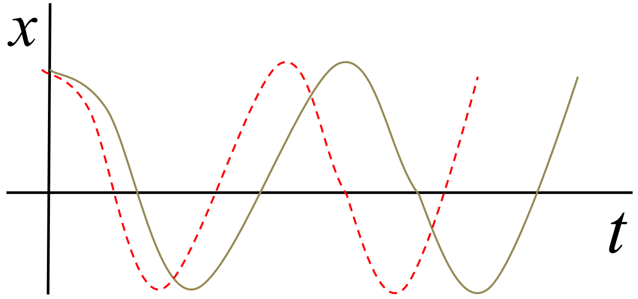
$$\ddot{x} + \omega_n^2 x = 0$$

Natural frequency $\omega_n = \sqrt{\frac{k}{m}}$

$$x(t) = c\cos(\omega_n t + \phi)$$

Frequency increases
with **stiffness**
and **inverse mass**





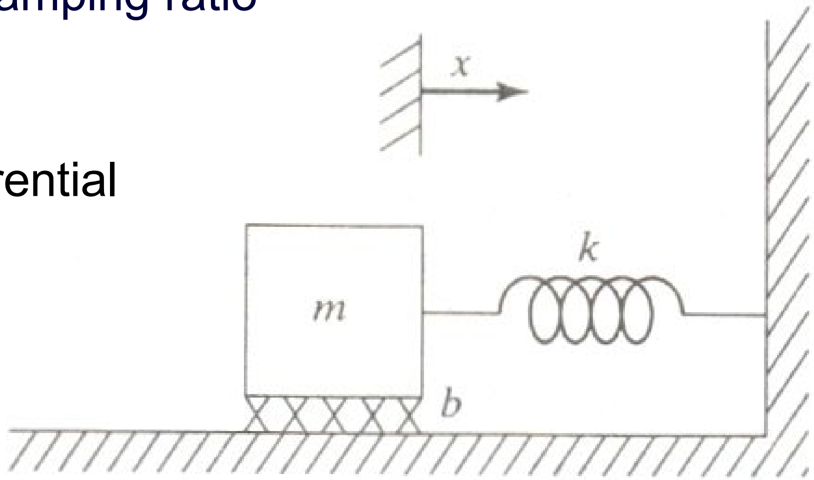
Spring-Mass System W/ Friction

$$m\ddot{x} + b\dot{x} + kx = 0, \quad \ddot{x} + \frac{b}{m}\dot{x} + \frac{k}{m}x = 0$$

$$\ddot{x} + 2\xi\omega_n\dot{x} + \omega_n^2 x = 0$$

$$\omega_n = \sqrt{\frac{k}{m}}$$

$$\xi = \frac{b}{2\omega_n m} = \frac{b}{2\sqrt{km}}$$ Natural damping ratio



Depending on the coefficient of the differential equation, we classify the responses into

① Underdamped
② Critically damped
③ Overdamped

Spring-Mass System W/ Friction

$$\ddot{x} + 2\xi\omega_n\dot{x} + \omega_n^2 x = 0$$

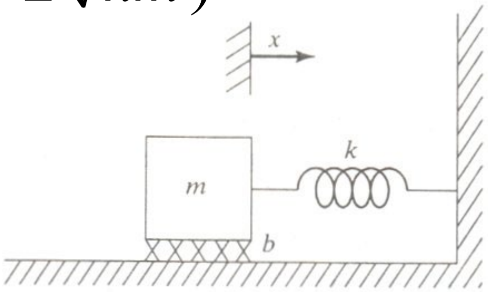
Underdamped system: $\xi < 1 \; (b < 2\sqrt{km})$

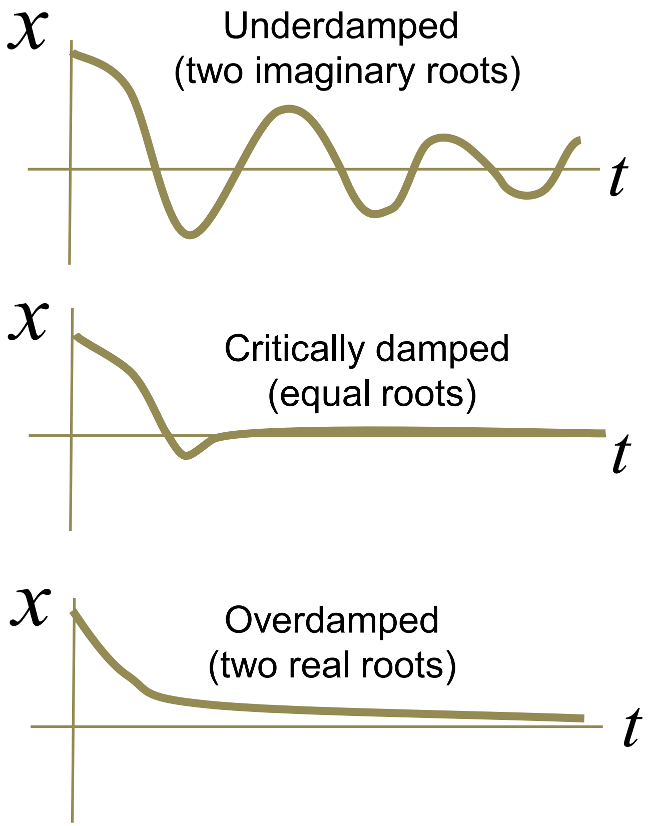
$$\boxed{x(t) = ce^{-\xi\omega_n t}\cos(\underbrace{\omega_n\sqrt{1-\xi^2}}_{\text{Damped natural frequency}}\, t + \phi)}$$

Damped natural frequency

Critically damped system:

$\xi = 1 \; (b = 2\sqrt{km})$

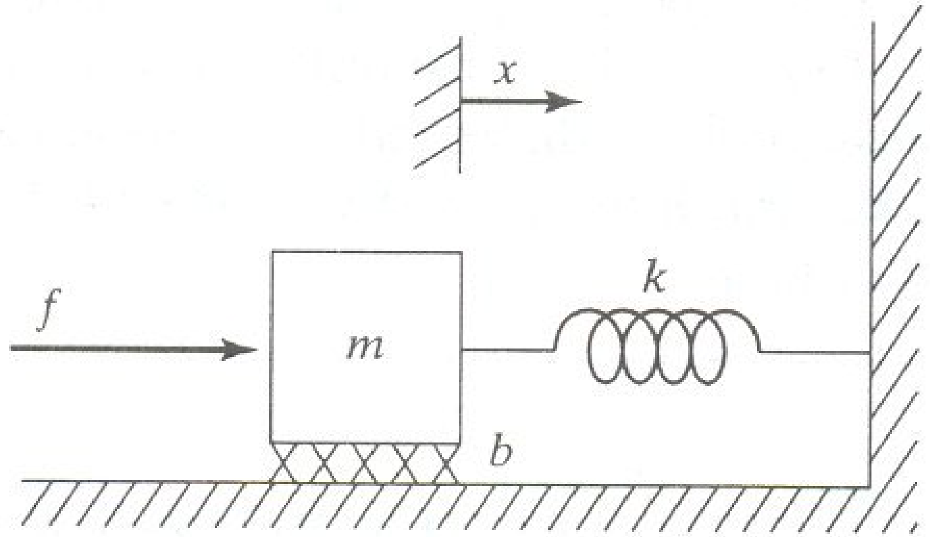




Control of Second Order System(1)

- Reshape the response of the system into the desired one and modify the system's behavior as desired.

$$m\ddot{x} + b\dot{x} + kx = f$$



Proportional-Derivative(PD) Control

Typical control law of the position regulation problem(PD control)

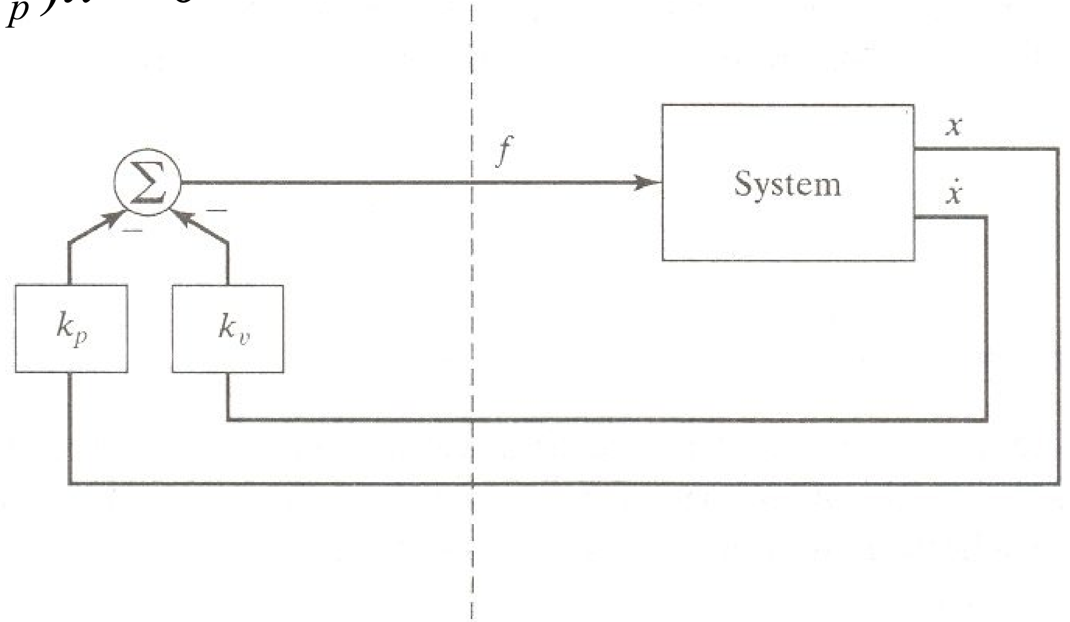
$$f = -k_p x - k_v \dot{x}$$

Closed loop dynamics will be

$$m\ddot{x} + b\dot{x} + kx = -k_p x - k_v \dot{x}$$

$$m\ddot{x} + (b + k_v)\dot{x} + (k + k_p)x = 0$$

$$m\ddot{x} + b'\dot{x} + k'x = 0$$



Control-law Partitioning(1)

Control law partitioning

- Controller = model-based portion + servo portion

Open loop equation of motion

$$m\ddot{x} + b\dot{x} + kx = f$$

Model based portion: to make use of estimated knowledge of m, b, k → **to reduce the system so that it appears to be a unit mass**, the input is shaped as

$$f = \alpha f' + \beta$$

Control-law Partitioning(2)

- Choose as follows.

$$\alpha = m$$

$$\beta = b\dot{x} + kx$$

- Substitute into the original model.

$$\ddot{x} = f'$$

  - It represents the dynamic equation of a unit mass.

Control-law Partitioning(3)

- In addition, the control law can be designed as

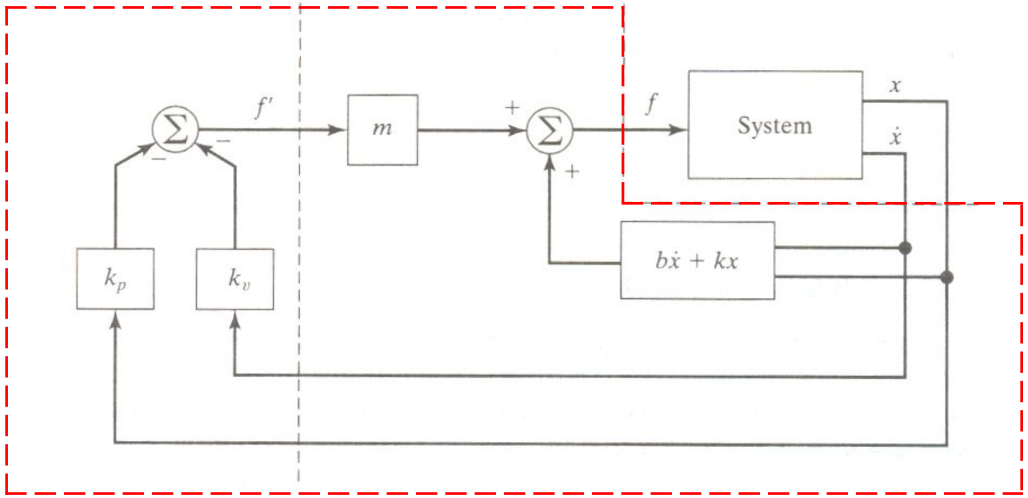
$$f' = -k_v \dot{x} - k_p x$$

- The overall dynamic equation will be

$$\ddot{x} + k_v \dot{x} + k_p x = 0$$

- Also, the control gain is selected to make it in critically damped state

$$k_v = 2\sqrt{k_p}$$



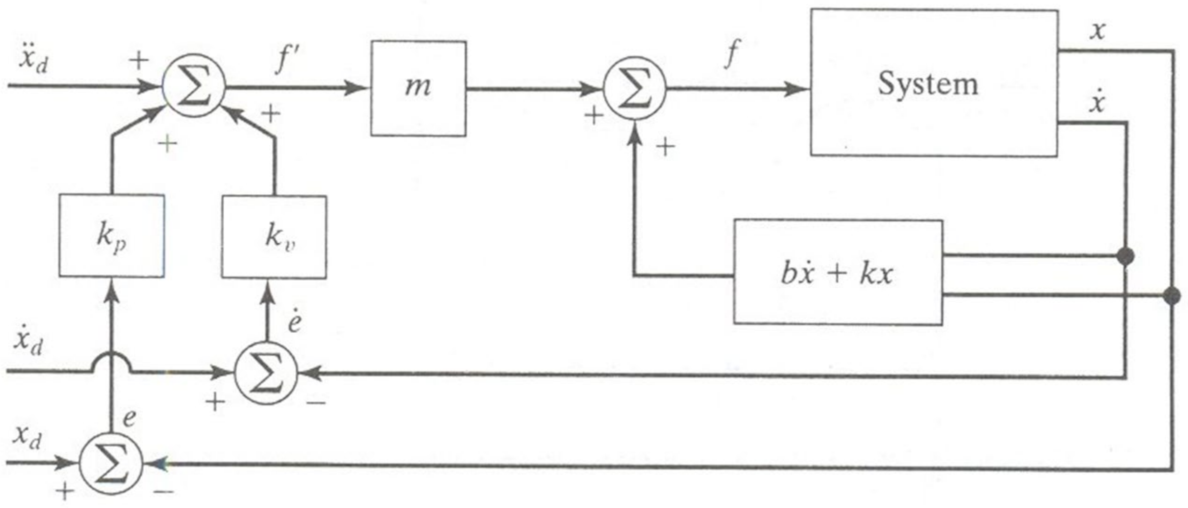
Trajectory Following Control(1)

- Trajectory as the function of time $x_d,\ \dot{x}_d,\ \ddot{x}_d$
- The servo control law for trajectory following is

$$f' = \ddot{x}_d + k_v \dot{e} + k_p e$$

- Total dynamic equation will be $e \equiv x_d - x$

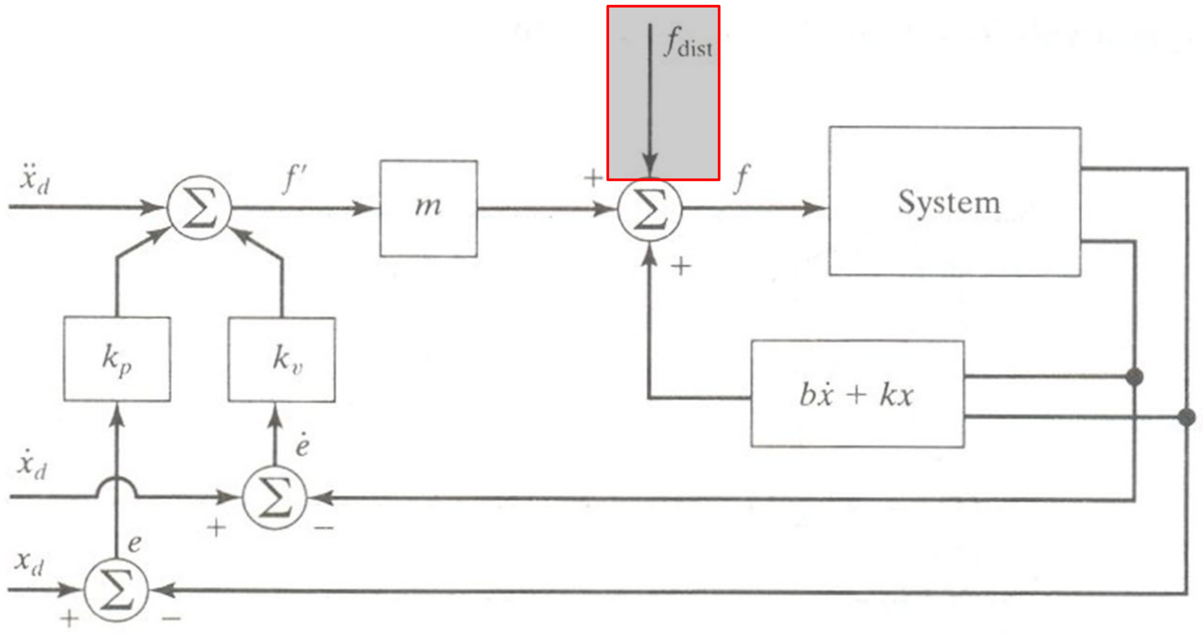
$$\ddot{e} + k_v \dot{e} + k_p e = 0$$



Disturbance Rejection(1)

- Bounded output under the bounded input

$$\ddot{e} + k_v \dot{e} + k_p e = f_{dist} \qquad \max_t f_{dist}(t) < a$$



Disturbance Rejection(2)

- Steady-state error

  (assuming that $f_{dist}(t) = const$)

  In the case of **steady state**, the derivative of all the system variables are zero

$$k_p e = f_{dist}$$

$$e = f_{dist} / k_p$$

$$\ddot{e} + k_v \dot{e} + k_p e = f_{dist}$$

As the position gain becomes large, the steady state error is getting smaller(but, never be zero)

Disturbance Rejection(3)

- Addition of an integral gain: to make the steady state error zero
  - PID control law: **proportional/integral/derivative**

$$f' = \ddot{x}_d + k_v \dot{e} + k_p e + k_i \int e dt$$

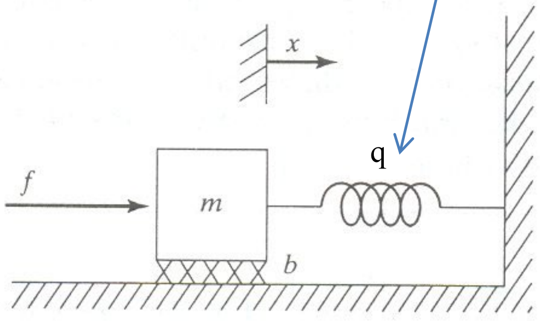
Error equation will be

$$\ddot{e} + k_v \dot{e} + k_p e + k_i \int e dt = f_{dist}$$

Example 10.1

- Keep the system critically damped

Nonlinear spring



Nonlinear spring

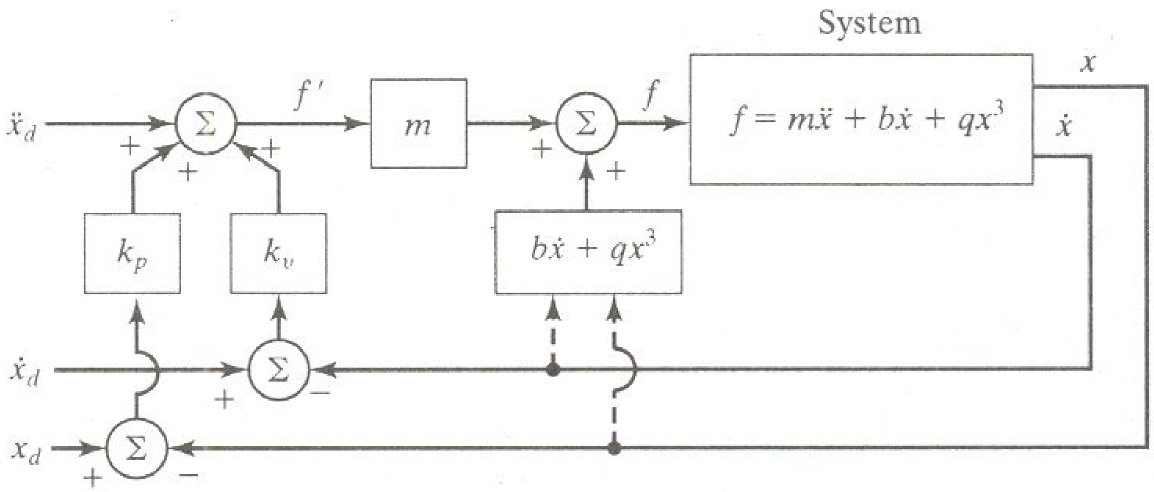
$$m\ddot{x} + b\dot{x} + qx^3 = f$$

Choose a controller

$$f = \alpha f' + \beta$$

$$\alpha = m$$

$$\beta = b\dot{x} + qx^3$$



$$\Rightarrow \ddot{x} = f'$$

$$f' = \ddot{x}_d + k_v \dot{e} + k_p e$$

Multi-input, Multi-output Control Systems

- Control law in MIMO system

$$\mathbf{F} = \alpha \mathbf{F}' + \boldsymbol{\beta}$$

Nonlinearity cancelling

- Servo law for a multidimensional system

$$\mathbf{F}' = \ddot{\mathbf{X}}_d + \mathbf{K}_v \dot{\mathbf{E}} + \mathbf{K}_p \mathbf{E}$$

Control Problem for Manipulators

- General manipulator dynamics

$$\boldsymbol{\tau} = \mathbf{M}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + \mathbf{V}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) + \mathbf{G}(\boldsymbol{\theta}) + \mathbf{F}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})$$

$$\boldsymbol{\tau} = \boldsymbol{\alpha}\boldsymbol{\tau}' + \boldsymbol{\beta}$$

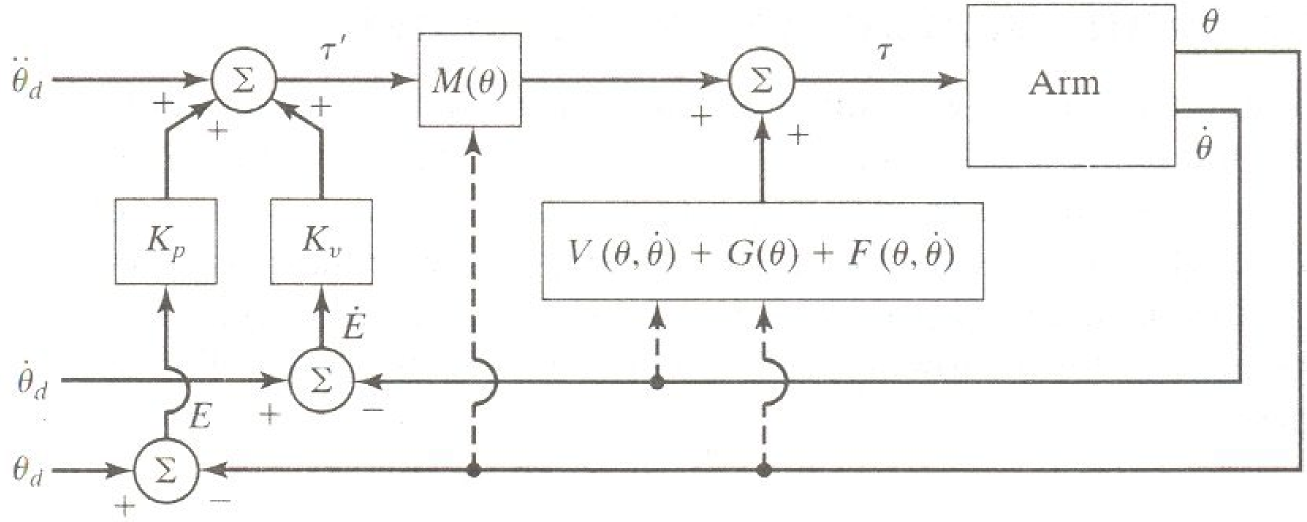
$$\boldsymbol{\alpha} = \mathbf{M}(\boldsymbol{\theta}) \quad \boldsymbol{\beta} = \mathbf{V}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) + \mathbf{G}(\boldsymbol{\theta}) + \mathbf{F}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})$$

$$\boldsymbol{\tau}' = \ddot{\boldsymbol{\theta}}_d + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_P\mathbf{E}$$

$$\ddot{\mathbf{E}} + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_P\mathbf{E} = \mathbf{0} \xrightarrow[\text{decoupling}]{} \ddot{e}_i + k_{vi}\dot{e} + k_{pi}e = 0$$

Practical Considerations

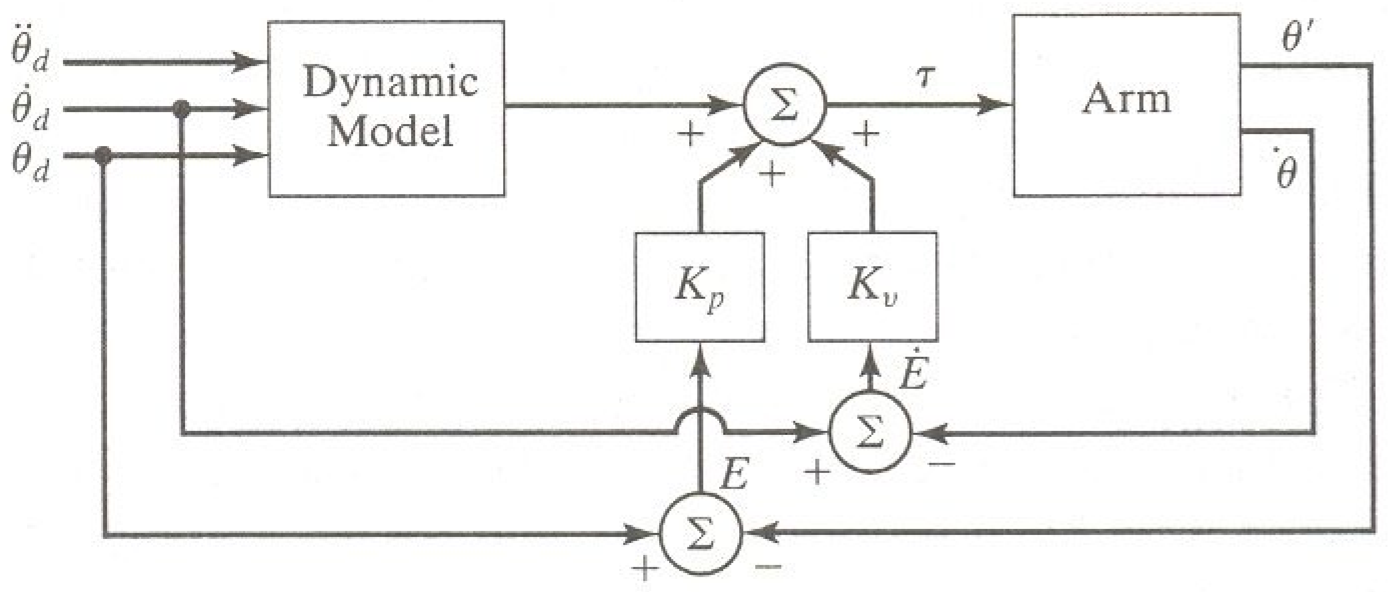
- Time required to compute the model
  - Sampling rate
  - Discrete-time control (actually not continuous time system !!!)



Feedforward Nonlinear Control

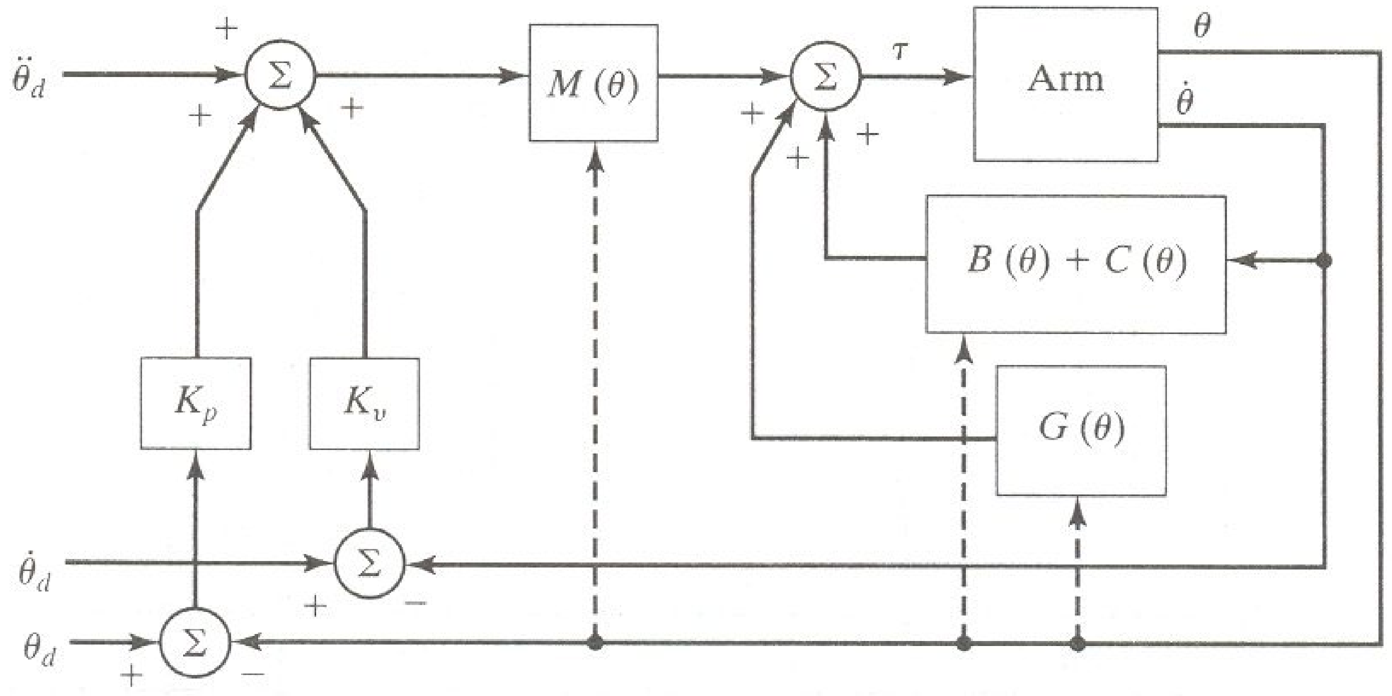
- Model-based torque added at a slower rate + fast inner servo loop
- Does not provide complete decoupling

$$\ddot{\mathbf{E}} + \mathbf{M}^{-1}(\mathbf{\Theta})\mathbf{K}_v\dot{\mathbf{E}} + \mathbf{M}^{-1}(\mathbf{\Theta})\mathbf{K}_P\mathbf{E} = \mathbf{0}$$



Dual-rate Computed Torque Implementation

- Possible practical implementation of decoupling and linearized position-control system
- Slow background process for computing dynamics
- Fast closed-loop servo



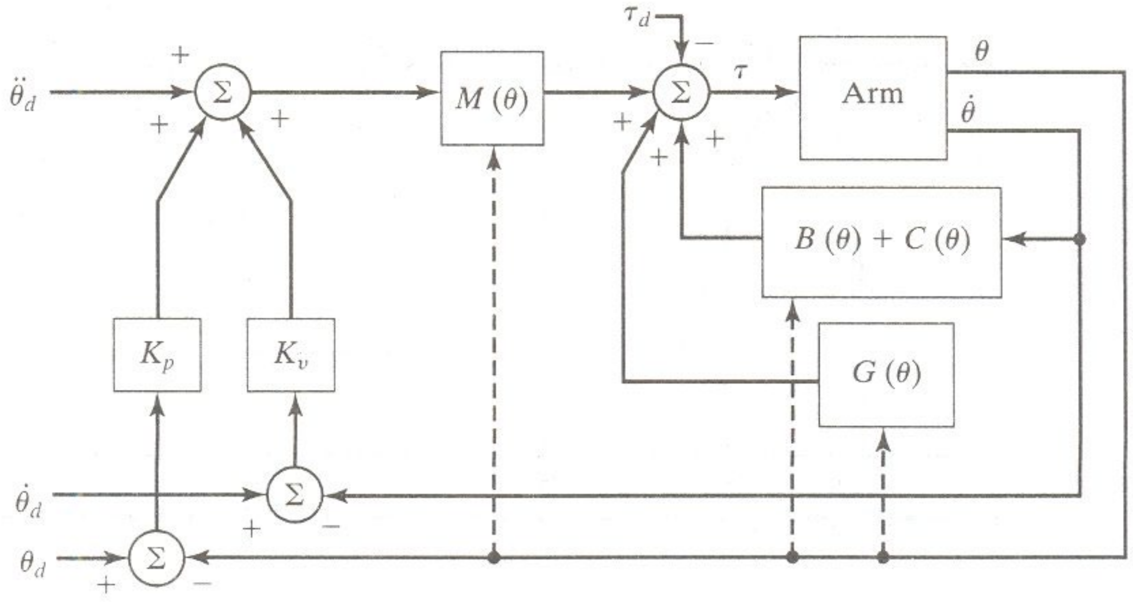
Lack of Knowledge of Parameters(1)

- Inaccurate manipulator dynamic model (e.g. friction)
- Effect of load → Change by grasping object
- Disturbances

$$\ddot{\mathbf{E}} + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_P\mathbf{E} = \mathbf{M}^{-1}(\boldsymbol{\Theta})\tau_d$$

steady-state error

$$\mathbf{E} = \mathbf{K}_p^{-1}\mathbf{M}^{-1}(\boldsymbol{\Theta})\tau_d$$



Lack of Knowledge of Parameters(2)

- Model parameter vs. Real parameter

$$\boldsymbol{\tau} = \mathbf{M}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + \mathbf{V}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) + \mathbf{G}(\boldsymbol{\theta}) + \mathbf{F}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})$$

$$\boldsymbol{\tau} = \boldsymbol{\alpha}\boldsymbol{\tau}' + \boldsymbol{\beta}$$

$$\boldsymbol{\alpha} = \hat{\mathbf{M}}(\boldsymbol{\theta})$$

$$\boldsymbol{\beta} = \hat{\mathbf{V}}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) + \hat{\mathbf{G}}(\boldsymbol{\theta}) + \hat{\mathbf{F}}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})$$

$$\ddot{\mathbf{E}} + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_P\mathbf{E} = \hat{\mathbf{M}}^{-1}[(\mathbf{M} - \hat{\mathbf{M}})\ddot{\boldsymbol{\theta}} + (\mathbf{V} - \hat{\mathbf{V}}) + (\mathbf{G} - \hat{\mathbf{G}}) + (\mathbf{F} - \hat{\mathbf{F}})]$$

Modeling error : not zero in general

Current Indusrial-Robot Control Systems

- Individual-joint PID control: each joint is controlled as a separate control system

$$\boldsymbol{\tau} = \boldsymbol{\alpha}\boldsymbol{\tau}' + \boldsymbol{\beta}$$

$$\boldsymbol{\alpha} = \mathbf{I}$$

$$\boldsymbol{\beta} = \mathbf{0}$$

$$\boldsymbol{\tau}' = \cancel{\ddot{\boldsymbol{\theta}}}_d + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_P\mathbf{E} + \mathbf{K}_i\int \mathbf{E}dt$$

Most cases : zero

- Addition of gravity compensation

$$\boldsymbol{\tau}' = \ddot{\boldsymbol{\theta}}_d + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_P\mathbf{E} + \mathbf{K}_i\int \mathbf{E}dt + \mathbf{G}(\boldsymbol{\theta})$$

Stability of Nonlinear System: Example

$$m\ddot{x} + b\dot{x} + kx = 0$$

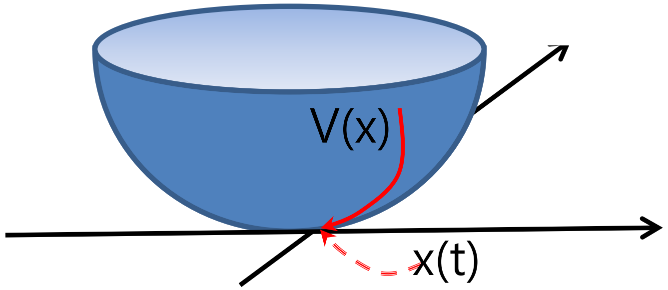
$$v = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$$

$$\dot{v} = m\dot{x}\ddot{x} + kx\dot{x} = -b\dot{x}^2 < 0$$

Lyapunov Stability

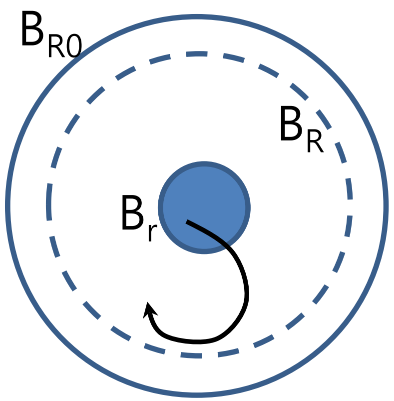
- Local stability
  - If, in a ball B_{R0}, there exists a scalar function V(x) with continuous first partial derivatives such that
    - V(x) is positive definite (locally in B_{R0})
    - dV/dt is negative semi-definite (locally in B_{R0})

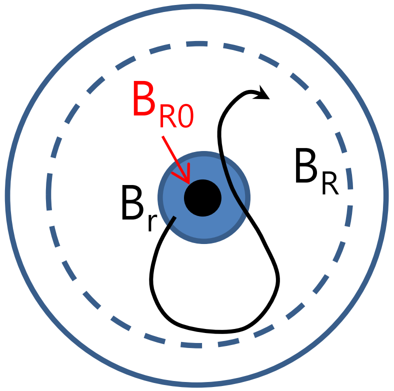
  then the equilibrium point 0 is stable. If, actually, the derivative dV(x)/dt is locally negative definite in B_{R0}, then the stability is asymptotic



Proof of Lyapunov Stability

- To show stability, we must show that given any strictly positive number *R*, there exists a smaller strictly positive number *r* such that any trajectory starting inside the ball B_r remains inside the ball B_R for all future time
- Negative definiteness of dV/dt
  - Asymptotic stability proof by contradiction
  - A trajectory starting in some ball Br
  - The trajectory will remain in BR for all future time
  - Since V is lower bounded and decreases continually, V tends towards a limit L, s. t. V(x(t))>L for all t>0.
  - Assume L > 0, then there exists Br0 that the trajectory never enters (from contradictory assumption)
  - Then, since –dV/dt is also continuous and positive definite, and since BR is bounded, -dV/dt must remain larger than some strictly positive number L1
  - This is contradiction, because it would imply that V(t) decreases from its initial value V0 to a value strictly smaller than L, in a finite time smaller than (V0-L)/L1
  - Thus, all trajectories starting in Br asymptotically converges





Lyapunov Stability Analysis(1)

- Stability analysis in nonlinear system
- Lyapunov stability analysis
  - Applicable to linear & nonlinear system
  - No implication on transient response/performance
  - No need to solve differential equation
- Definition
  - Xe is stable if starting close enough to Xe at t0, the state will always stay close to Xe at later times.
  - I.E. Xe is Lyapunov stable if for any given $e>0$, there exists a $d>0$ such that if $\|X0-Xe\|<d(e,t0)$, then $\|X(t)-Xe\|<e$ for all $t \geq t0$

Lyapunov Stability Analysis(2)

- Stability of differential equation

$$\dot{\mathbf{X}} = f(\mathbf{X})$$

- $v(\mathbf{X})$ has continuous first partial derivative, $\dot{v}(\mathbf{X}) \le 0$ for all X except

$$v(0) > 0 \qquad v(0) = 0$$

-

- Linear system $\dot{x} = Ax$
  - What is the stability condition for this linear system?
  - Control Lyapunov function

$$V(x) = \frac{1}{2}x^T x \Rightarrow \dot{V}(x) =$$

Remind : Classical Passivity-based Control

- No contact, no friction

$$M(\theta)\ddot{\theta} + C(\theta,\dot{\theta})\dot{\theta} + r(\dot{\theta}) + G(\theta) = \tau$$

- Consider error between the actual and the desired link position
- Set-point control $\tilde{\theta} = \theta - \theta_d$
- Control Lyapunov Function $[\theta_d, 0]^T$

Additional damping

Choose a controller

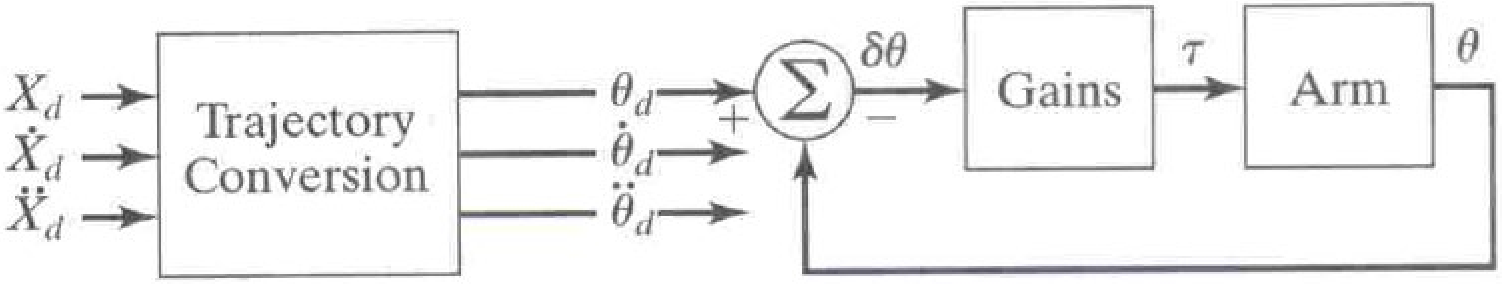
$$\tau = \underbrace{G(\theta)}_{\text{Gravity compensation}} - \underbrace{K_P\tilde{\theta}}_{\text{Stiffness control}} - \underbrace{K_D\dot{\theta}}_{\text{Additional damping}}$$

$$H(\tilde{\theta},\dot{\theta}) = \frac{1}{2}\dot{\theta}^T M(\theta)\dot{\theta} + \frac{1}{2}\tilde{\theta}^T K_P\tilde{\theta}$$

$$\begin{aligned}\dot{H} &= \dot{\theta}^T M(\theta)\ddot{\theta} + \frac{1}{2}\dot{\theta}^T \dot{M}(\theta)\dot{\theta} + \tilde{\theta}^T K_P\dot{\theta} \\ &= \frac{1}{2}\dot{\theta}^T \underbrace{\left[\dot{M} - 2C\right]}_{\text{0: skew-symmetric}}\dot{\theta} - \underbrace{\dot{\theta}^T r(\dot{\theta})}_{\text{Damping term} \geq 0} + \dot{\theta}^T\left[\tau - G(\theta) + K_P\tilde{\theta}\right] \leq 0\end{aligned}$$

Cartesian-based control

- Joint-based control with Cartesian-path input



$$\theta_d = INVKIN(x_d)$$

$$\dot{\theta}_d = J^{-1}(\theta)\dot{x}_d$$

$$\ddot{\theta}_d = \dot{J}^{-1}(\theta)\dot{x}_d + J^{-1}\ddot{x}_d$$

Computationally heavy

or

$$\theta_d = INVKIN(x_d)$$

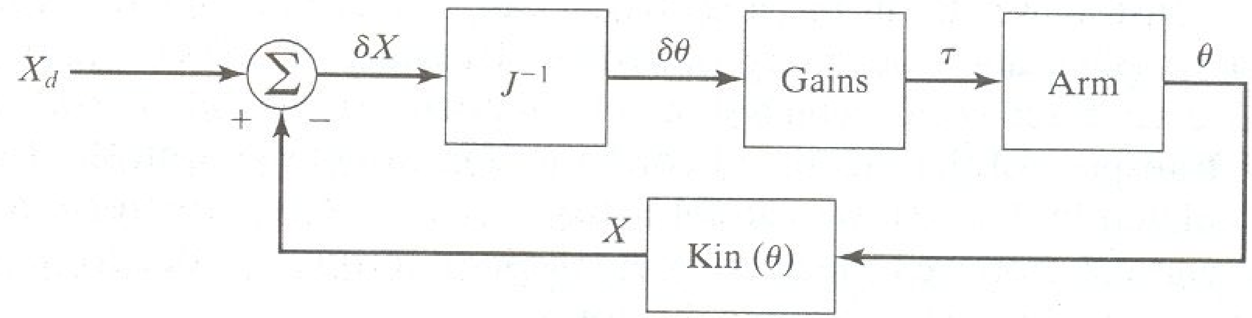
$$\dot{\theta}_d = \frac{\theta_d(k) - \theta_d(k-1)}{dT}$$

$$\ddot{\theta}_d = \frac{\dot{\theta}_d(k) - \dot{\theta}_d(k-1)}{dT}$$

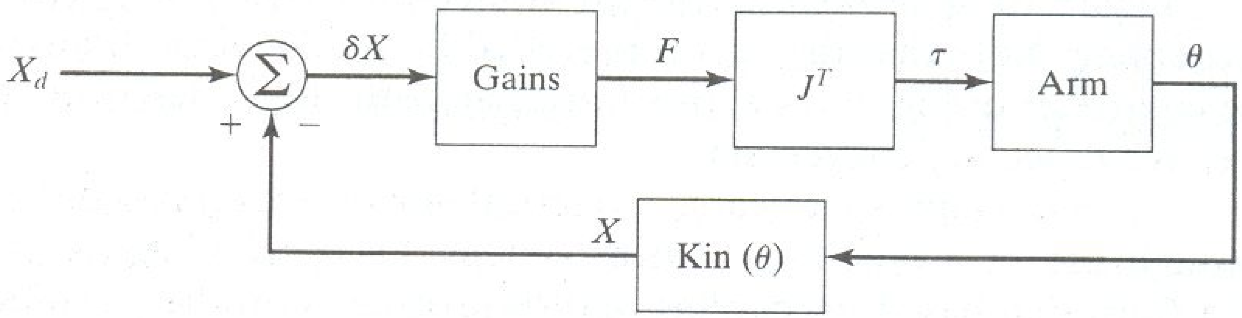
Sensitive to noise

Cartesian-based Control Systems

- Inverse-Jacobian controller

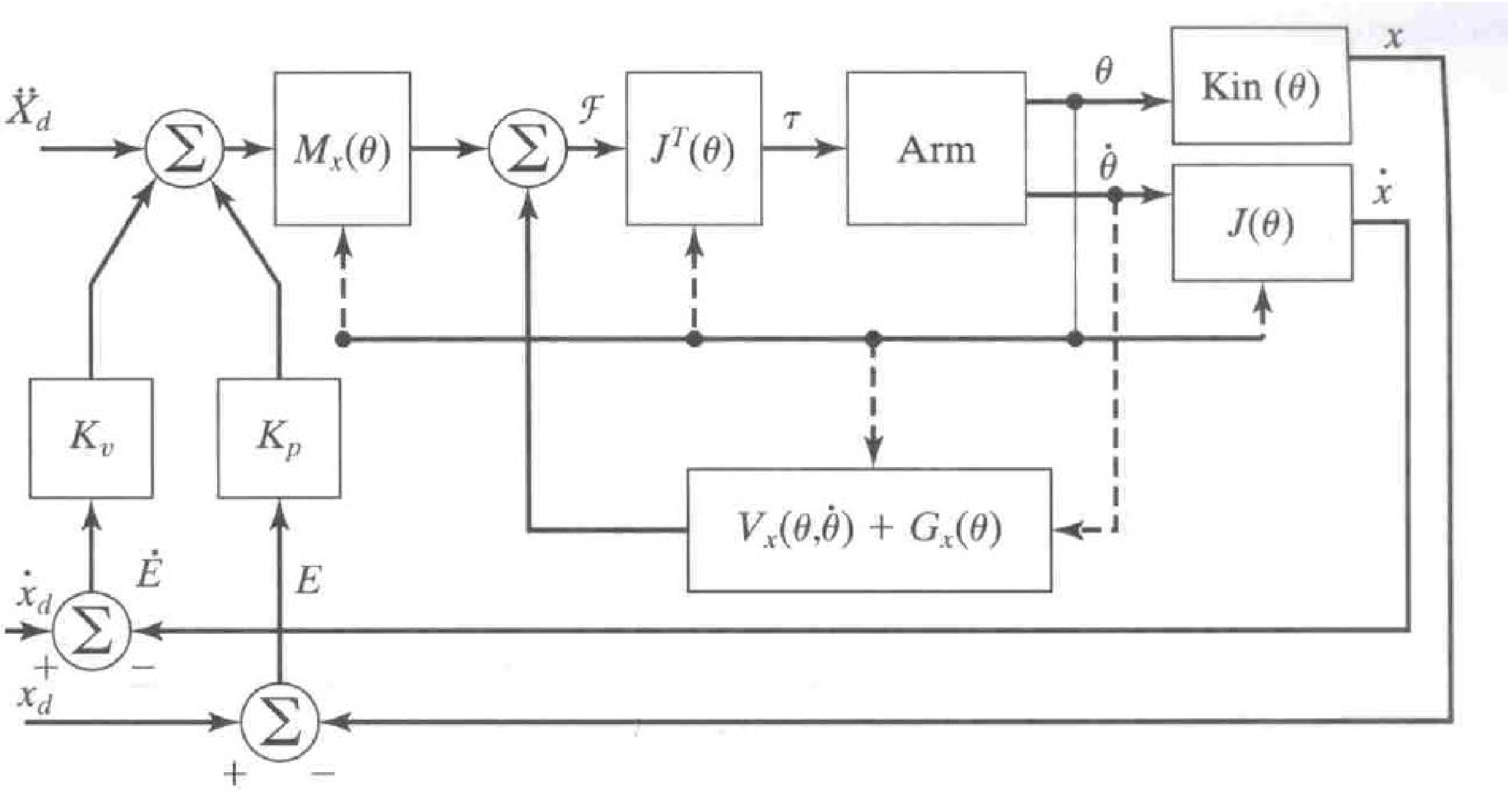


- Transpose Jacobian Cartesian controller

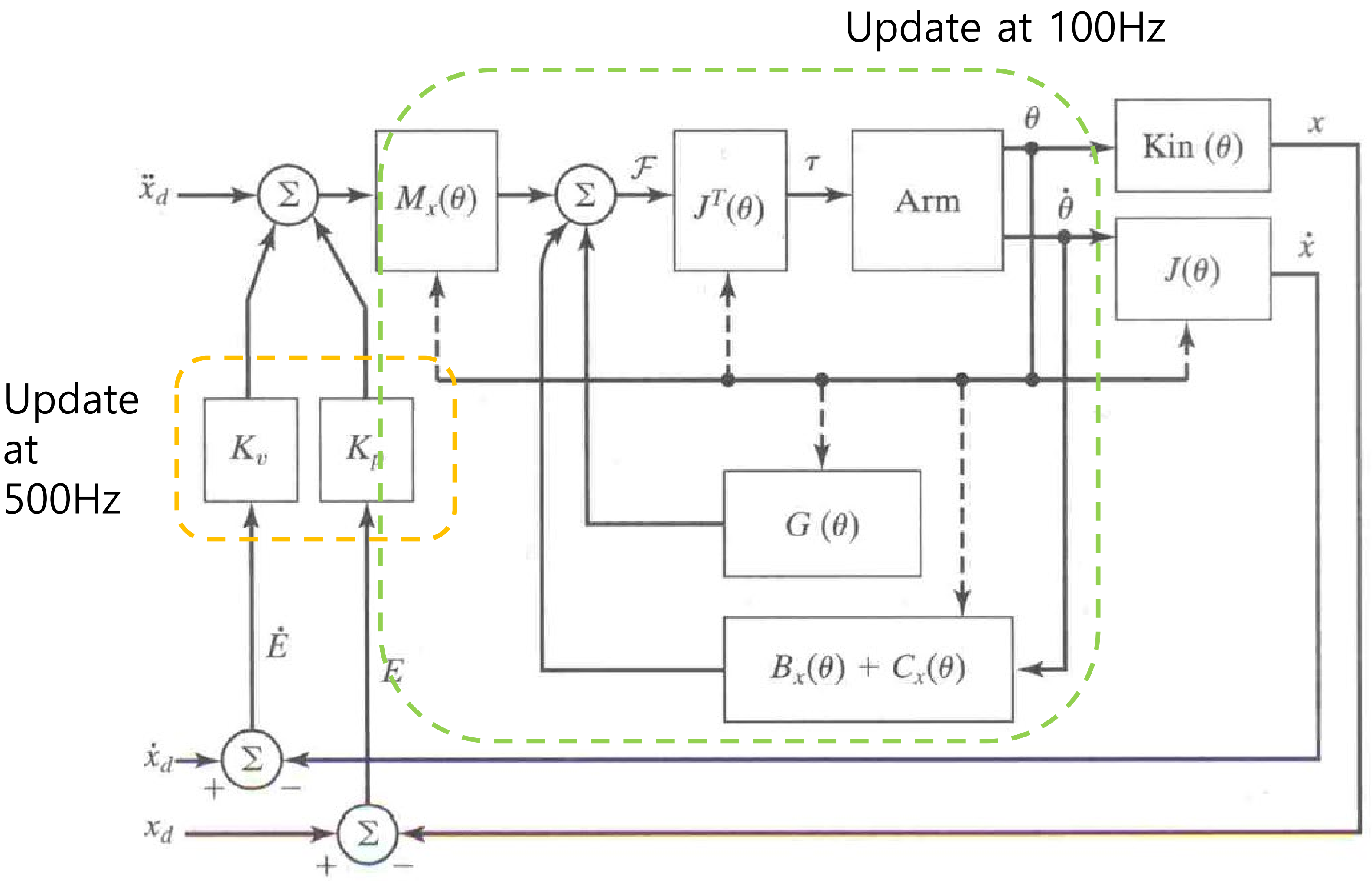


No fixed gain satisfies all configuration

Cartesian model-based control



Implementation of Cartesian model-based control

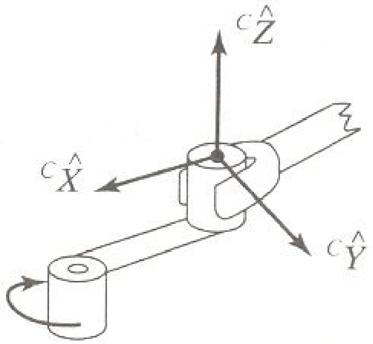


Force Control: Introduction

- Specify a force rather than position
- Hybrid position/force controller
- Still, not be applied in general
- In case of part-mating, force control is prerequisite
- Position and force are orthogonal spaces

A Framework for Control in Partially Constrained Tasks(1)

- Natural constraint(position, force)



Natural constraints

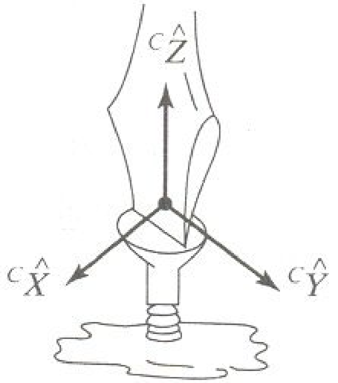
$v_x = 0$ $\quad f_y = 0$

$v_z = 0$ $\quad n_z = 0$

$v_x = 0$

$\omega_y = 0$

(a) Turning crank



Natural constraints

$v_x = 0$ $\quad f_y = 0$

$\omega_x = 0$ $\quad n_z = 0$

$\omega_y = 0$

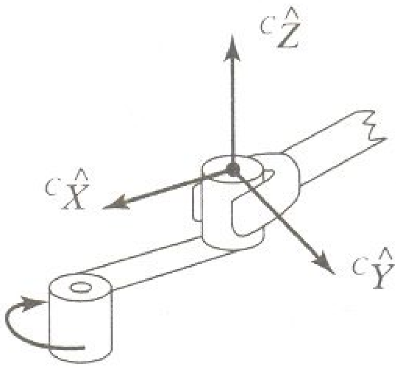
$v_z = 0$

(b) Turning screwdriver

A Framework for Control in Partially Constrained Tasks(2)

- Artificial constraint
- Assembly strategy: planned artificial constraints
- Task analysis is necessary

(a) Turning crank



Natural constraints

$v_x = 0$ $\quad f_y = 0$

$v_z = 0$ $\quad n_z = 0$

$\omega_x = 0$

$\omega_y = 0$

Artificial constraints

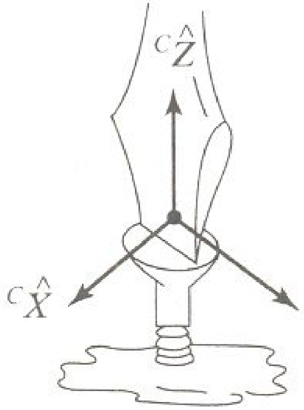
$v_y = r\alpha_1$ $\quad f_x = 0$

$\omega_z = \alpha_1$ $\quad f_z = 0$

$n_x = 0$

$n_y = 0$

(b) Turning screwdriver



Natural constraints

$v_x = 0$ $\quad f_y = 0$

$\omega_x = 0$ $\quad n_z = 0$

$\omega_y = 0$

$v_z = 0$

Artificial constraints

$v_y = 0$ $\quad f_x = 0$

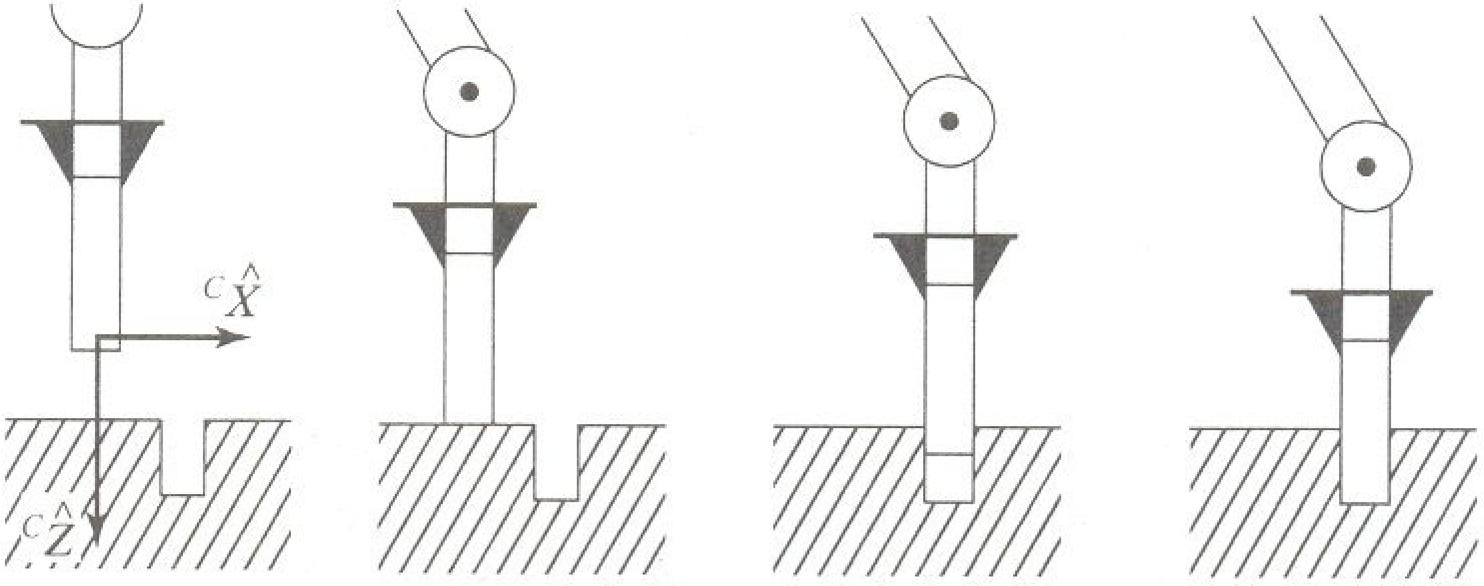
$\omega_z = \alpha_2$ $\quad n_x = 0$

$n_y = 0$

$f_z = \alpha_3$

Ex 11.1

- Task sequences



(a) (b) (c) (d)

Force Control of Mass-Spring System(1)

$$f_e = k_e x$$

$$f = m\ddot{x} + k_e x + f_{dist}$$

$$f = mk_e^{-1}\ddot{f}_e + f_e + f_{dist}$$

$$\alpha = mk_e^{-1}$$

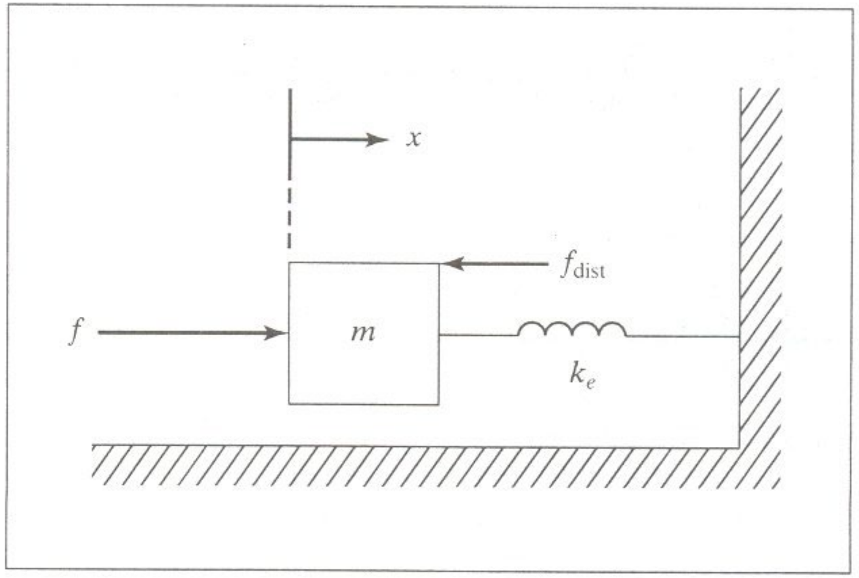
$$\beta = f_e + f_{dist}$$

Choose our control

$$f = mk_e^{-1}[\ddot{f}_d + k_{vf}\dot{e}_f + k_{pf}e_f] + f_e + f_{dist}$$

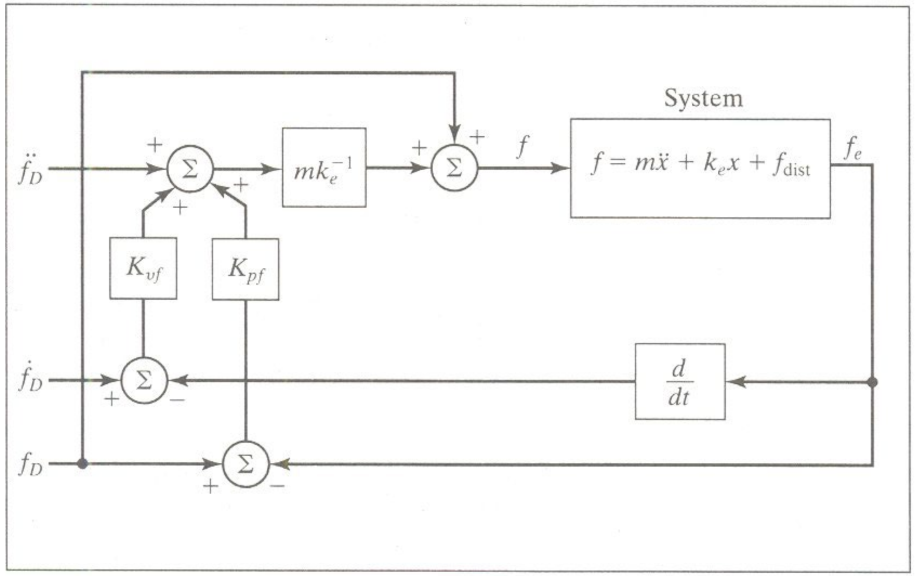
Closed loop system

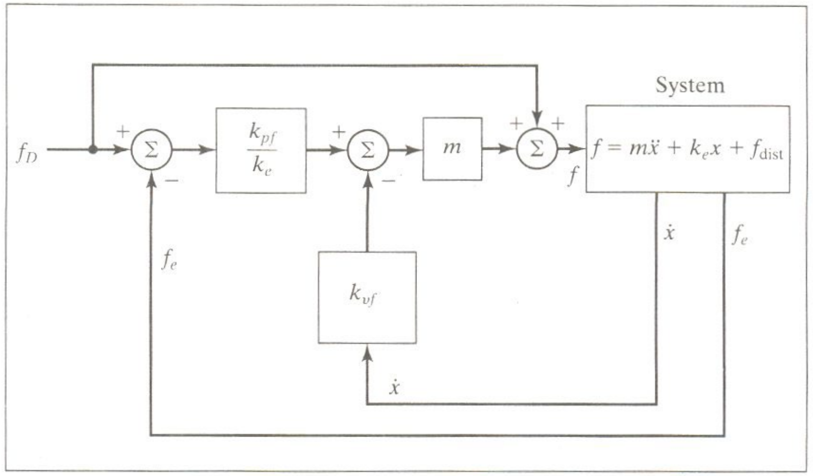
$$\ddot{e}_f + k_{vf}\dot{e}_f + k_{pf}e_f = 0$$



Force Control of Mass-Spring System(2)

$$f = mk_e^{-1}[\ddot{f}_d + k_{vf}\dot{e}_f + k_{pf}e_f] + f_d$$



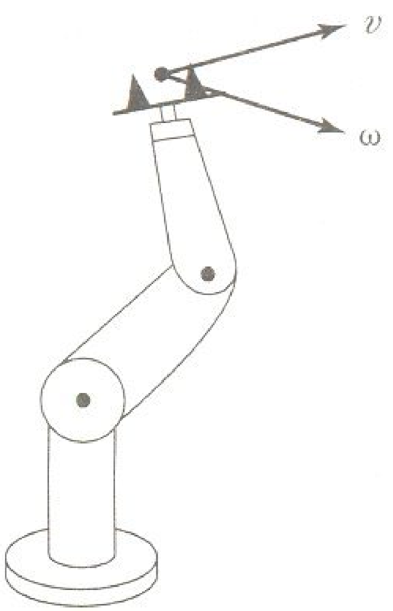


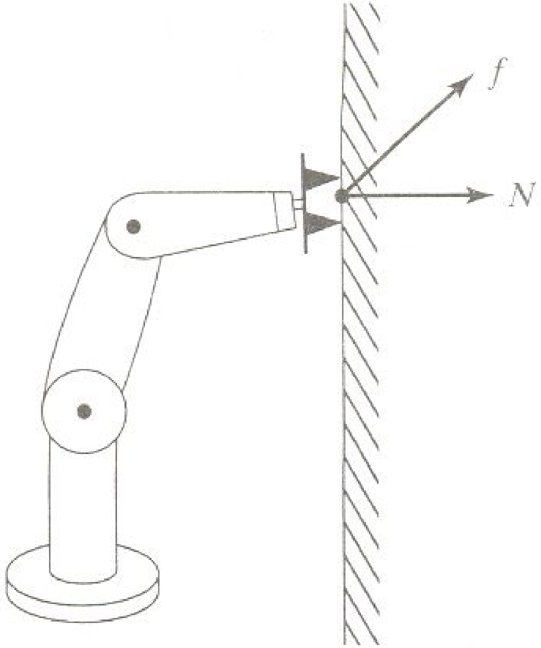
When fd = const

$$f = m[k_{pf}k_e^{-1}e_f - k_{vf}\dot{x}] + f_d$$

Hybrid Position/Force Control Problem

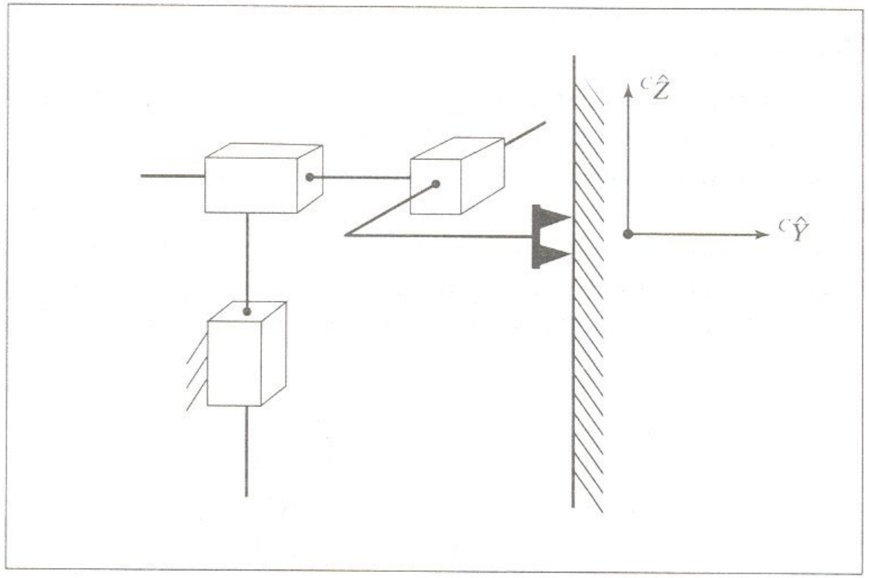
- Position control along the directions in which a natural force constraint exists
- Force control along the direction in which a natural position constraint exists

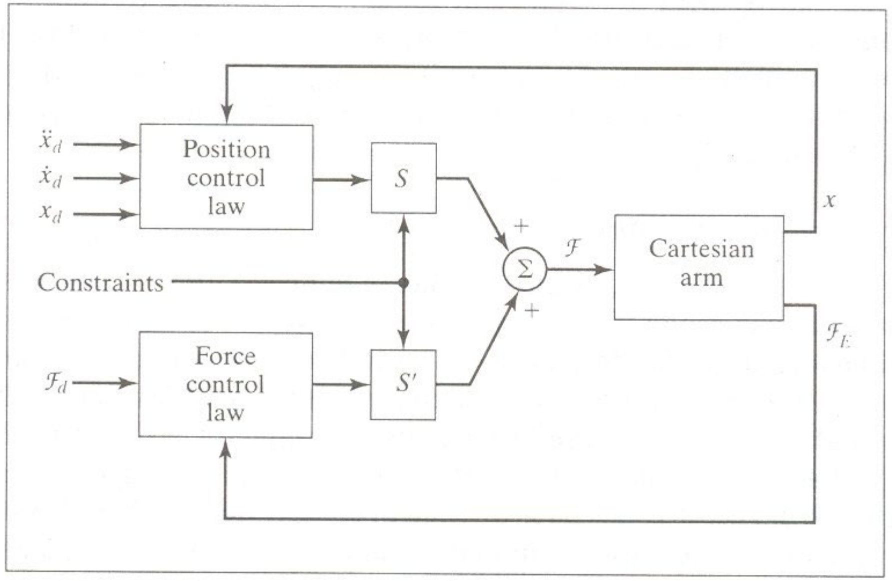




Hybrid Position/Force Control Scheme(1)

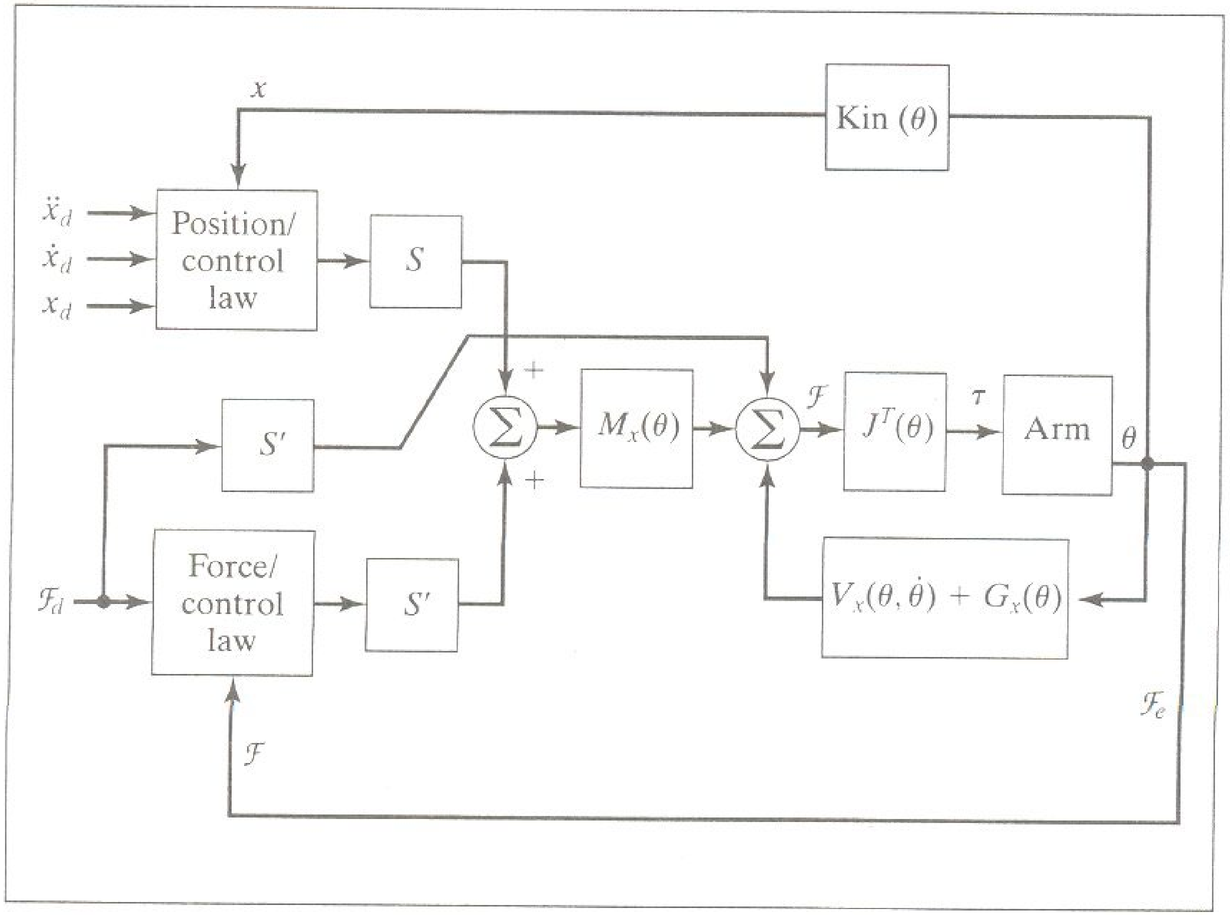
- Cartesian Manipulator Aligned with {C}





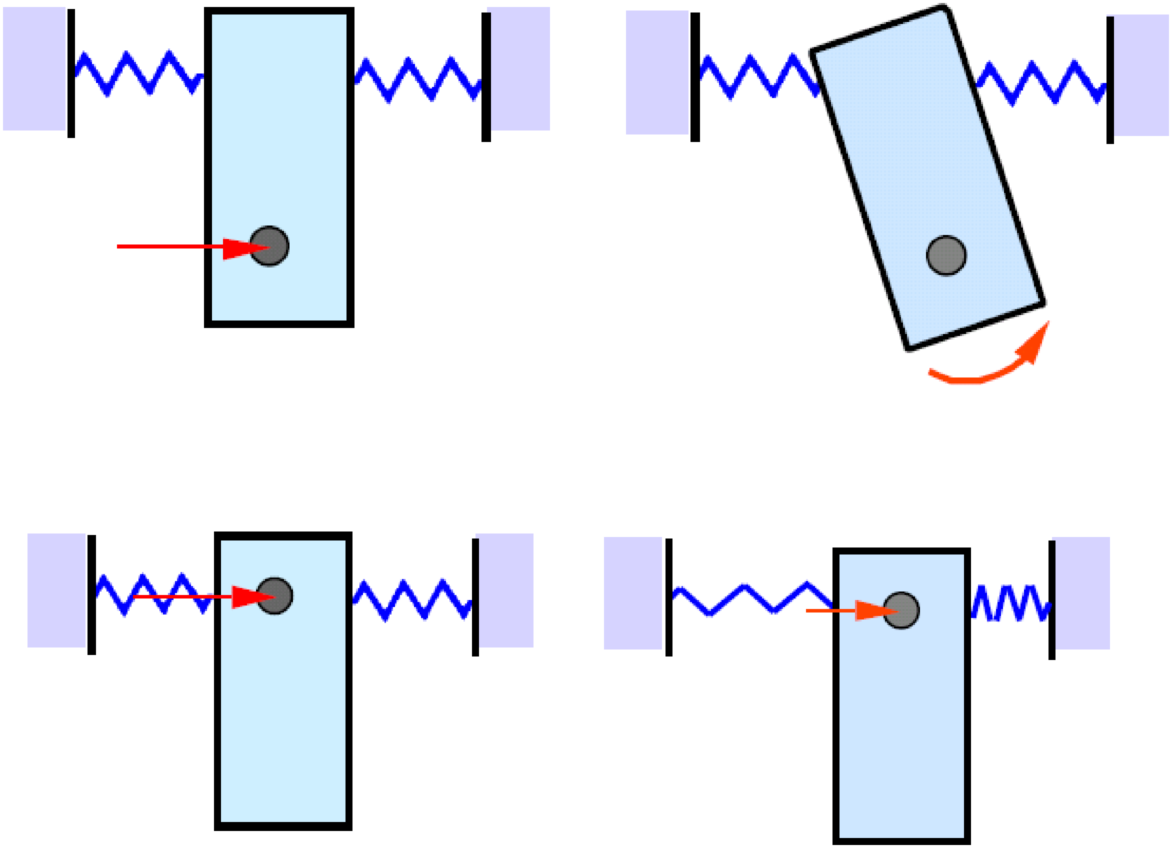
Hybrid Position/Force Control Scheme(2)

- General manipulator



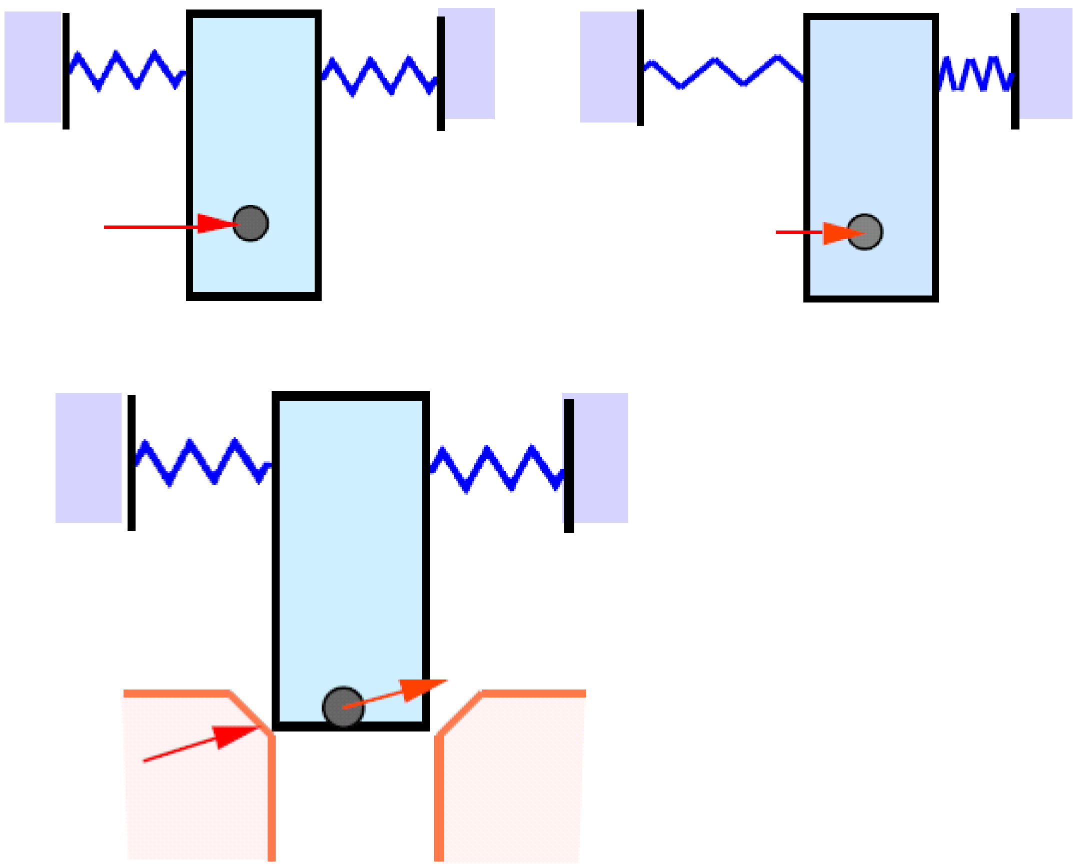
Current Industrial-Robot Control Systems

- Passive compliance



Remote Center of Compliance(RCC)

- Peg-in-Hole Problem



video

Compliance Through Softening Position Gains

- Active stiffness control(by Salisbury)
  - Ex) Grinding

$$\mathbf{F} = \mathbf{K}_{px}\delta\mathbf{X}$$ desired spring behavior (in Cartesian space)

$$\delta\mathbf{X} = \mathbf{J}(\boldsymbol{\theta})\delta\boldsymbol{\theta}$$

$$\mathbf{F} = \mathbf{K}_{px}\mathbf{J}(\boldsymbol{\theta})\delta\boldsymbol{\theta}$$

$$\boldsymbol{\tau} = \mathbf{J}^T\mathbf{F} = \mathbf{J}^T(\boldsymbol{\theta})\mathbf{K}_{px}\mathbf{J}(\boldsymbol{\theta})\delta\boldsymbol{\theta}$$ Joint control law

Joint-based position control law $\tau = K_P E + K_v \dot{E}$

Joint-based position control law $\tau = J^T(\Theta) K_{P_x} J(\Theta) E + K_v \dot{E}$